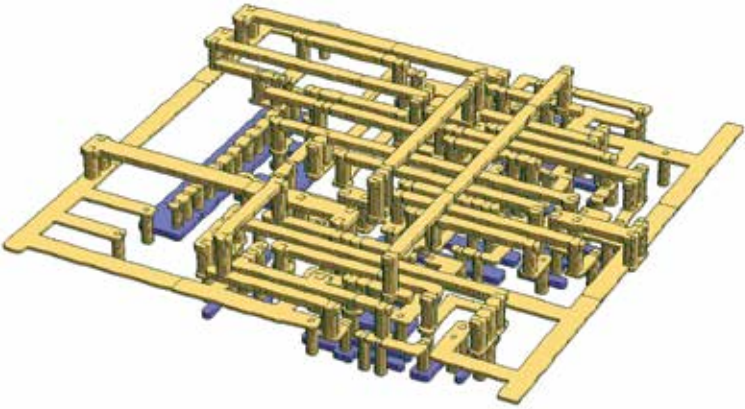




One smart methodology

The Boolean logic function of the cell is called its logical view: the functional behavior gets captured as Boolean algebra equation or a truth table. In some cases, it's captured as a state transition table.

In the normal course, the primary design of a standard cell gets developed at the transistor level. It appears as a schematic view of a transistor netlist. The netlist refers to a nodal description of the transistors, which detail the transistors' connections with each other and also that of the terminal ports with the external environment. A schematic view could be produced using different Electronic Design Automation(EDA) and Computer Aided Design(CAD) programs. These programs generally come with a Graphical User Interface(GUI) that aids the generation of this netlist.

Additional CAD programs could also be used by designers-like Spectre of SPICE which could effectively simulate the netlist's electronic behavior. This is done by declaring the input stimulus(current waveforms or voltage) following which the circuit's time domain or analogue response is calculated. Using the simulations, it can be verified if the netlist implements the required function and also predict other relevant parameters. Examples of such parameters include signal propagation delay and power consumption.

It should be noted that the logical as well as netlist views are useful only for abstract or algebraic simulation, but not for device fabrication. This means that the standard cell's actual physical representation must also be designed. As far as common design practices go, this view- called the layout view is the lowest level of design abstraction.